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1 Inclusion-Exclusion

2 sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

3 sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

4 sets

$$\begin{aligned} |A \cup B \cup C \cup D| = & |A| + |B| + |C| + |D| \\ & - |A \cap B| - |A \cap C| - |A \cap D| - |B \cap C| - |B \cap D| - |C \cap D| \\ & + |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| \\ & - |A \cap B \cap C \cap D| \end{aligned}$$

Generally for n sets

If $A_1, A_2, \dots, A_n$ are n finites sets, then:

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| = & \sum_{1 \leq i \leq n} |A_i| \\ & - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ & + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| \\ & - \dots \\ & + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \end{aligned}$$

1.1 – Inclusion-Exclusion of properties

For x elements we have n properties $P_1, P_2, \dots, P_n$. We say that P'_i is the negation of property P_i .

The number of elements satisfying all properties P_i, P_j, P_k is $N(P_i, P_j, P_k)$

The number of elements that satisfy *none* of the properties is:

$$\begin{aligned} N(P'_1, P'_2, \dots, P'_n) = & N - \sum_{1 \leq i \leq n} N(P_i) \\ & + \sum_{1 \leq i < j \leq n} N(P_i, P_j) \\ & - \sum_{1 \leq i < j < k \leq n} N(P_i, P_j, P_k) \\ & + \dots \\ & + (-1)^n N(P_1, P_2, \dots, P_n) \end{aligned}$$

where N is the total number of elements.

1.2 – Derangements

A derangement is a permutation that leaves no object in its original position. The amount of derangements of n objects, denoted by $!n$, is given by:

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right]$$

1.3 – Binomial expanesions

The binomial expansion of $(x + y)^n$ is given by:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

Example with coefficients:

$$(2x^2 - 3y^3)^8 = \sum_{k=0}^8 \binom{8}{k} (2x^2)^{8-k} (-3y^3)^k$$

10 Graphs

Table 1: Graph Terminology

Type	Edges	Allow Multiple Edges?	Allow Loops?
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Both	Yes	Yes

Multiple edge mean multiple edge between the same pair of vertices. A loop is an edge that connects a vertex to itself.

10.1 – Degrees and vertices

Sum(degrees) = 2 * Number of edges = even

Count(odd degree vertices) = even

Sum(in-degrees) = Sum(out-degrees) = Number of edges

10.2 – Common Graphs

Complete Graphs K_n

Graph with one edge between each pair of vertices.

Vertices: n

Edges: $\frac{n(n-1)}{2}$

Degrees of vertices: $n - 1$

Cycles C_n

Graph where the vertices are connected in a closed chain. Only defined for $n \geq 3$.

Vertices: n

Edges: n

Degrees of vertices: 2

Wheels W_n

A cycle with an additional center vertex connected to all other vertices. Only defined for $n \geq 4$.

Vertices: n

Edges: $2(n - 1)$

Degrees of vertices: Center: $n - 1$, Other: 3

n-cubes Q_n

A cube of dimension n . A cube of dimension n can be made by connecting the corresponding vertices of two cubes of dimension $n - 1$.

Vertices: 2^n

Edges: $n2^{n-1}$

Degrees of vertices: n

10.3 – Bipartite Graphs

A graph that can be split in two subsets, where no items in the same subsets are connected to each other.

All elements in one subset do not need to be connected to all elements in the other subset.

An easy test if a graph is bipartite is to try to color the graph with two colours, where no two connected vertices have the same color.

Properties of common graphs:

Complete graphs are not bipartite for $n > 2$.

Cycles are bipartite if they have an even number of vertices.

Wheels are not bipartite.

n-cubes are bipartite.

Complete bipartite graphs

A bipartite graph where all vertices in one subset are connected to all vertices in the other subset. Denoted $K_{m,n}$, where m and n are the number of vertices in each subset.

Matchings

A matching M in a graph G is a subset of edges such that no two edges in M share a common vertex. No vertex is neighbour to more than one vertex in the matching.

Vertices at the ends of edges in the matching are said to be *matched*, while vertices not connected to any edge in the matching are *unmatched*.

A *Maximum matching* is a matching that contains the largest possible number of edges.

A matching in a bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) is a *complete matching* from V_1 to V_2 if every vertex in V_1 is connected.

Do note: Not every element in V_2 needs to be connected.

Necessary and sufficient conditions for complete matching

Hall's Marriage Theorem

The bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) has a complete matching from V_1 to $V_2 \iff \forall S \subseteq V_1 : |N(S)| \geq |S|$, where $N(S)$ is the set of all neighbours of vertices in S .

10.4 – New graphs

Subgraphs

Graph $H = (W, F)$ is a subgraph of $G = (V, E)$ if $W \subseteq V$ and $F \subseteq E$.

H is a *proper subgraph* of G if $H \neq G$.

Given graph $G = (V, E)$, an *induced subgraph* is a subgraph made by taking all vertices of a subset $W \subseteq V$ and all edges in E that connect to vertices in W on both ends.

Adding and removing Edges

The subgraph induced by removing edge e from G is denoted by $G - e = (V, E - \{e\})$. Likewise, an edge e can be added to G to form $G + e = (V, E \cup \{e\})$.

Edge Contractions

The contraction of edge $e = \{u, v\}$ in graph G is denoted by G/e . It is formed by removing u and v from G , adding a new vertex w , and connecting all neighbours of u and v to w .

Do note: This is not a subgraph of G .

Removing vertices

The subgraph induced by removing vertex v from G is denoted by $G - v = (V - \{v\}, E')$, where E' is the set of edges in E that do not connect to v .

Graph unions

The union of graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is denoted by $G_1 \cup G_2 = (V_1 \cup V_2, E_1 \cup E_2)$. Includes all vertices and edges from both graphs, does not double up on vertices or edges that are in both graphs.

10.5 – Graph representation

Adjacency lists

Graphs can be represented as adjacency lists. Example:

Vertex	Neighbours
A	B, C
B	A, D
C	A, D
D	B, C

Table 2: Adjacency list of undirected graph A

A directed graph can also be represented with an initial and terminal column. Example (not the same graph as above):

Initial Vertex	Terminal Vertex
A	A,B,C,D
B	C,D
C	-
D	A,C,D

Table 3: Adjacency list of directed graph B

Adjacency matrices

Graphs can also be represented as adjacency matrices. Given graph G with n vertices, the adjacency matrix of G A_G is an $n \times n$ matrix where for entry a_{ij} , it is 1 if there is an edge from vertex i to vertex j , and 0 otherwise. Example:

$$A_G = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Table 4: Adjacency matrix of undirected graph A

This is the same as the vertex-neighbour representation above.

Do note: There is often no objective way to number the vertices, so adjacency matrices can vary for the same graph. (up to $n!$ matrices)

Adjacency matrices can also represent multigraphs and pseudographs, where the entries represent the amount of edges. The diagonal represents the amount of loops. Example:

$$A_G = \begin{bmatrix} 2 & 3 & 1 & 0 \\ 3 & 0 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

Table 5: Adjacency matrix of a pseudograph C

Here vertex 1 has 2 loops, 3 edges to vertex 2 and 1 edge to vertex 3.

Symmetry: Undirected graphs have symmetric adjacency matrices, directed graphs might not.

Graphs for directed graphs are the same, but a_{ij} represents an edge from vertex i to vertex j . Example:

$$A_G = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Table 6: Adjacency matrix of directed graph B

Do note: If v_1 connects to v_2 , then row 1, column 2 is 1.

Incidence matrices

An incidence matrix represents the connections between vertices and edges. Given graph G with n vertices and m edges, the incidence matrix of G I_G is an $n \times m$ matrix where for entry i_{ij} , it is 1 if vertex i is connected to edge j , and 0 otherwise. Example:

$$I_G = \begin{bmatrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ v_1 & 1 & 1 & 0 & 0 & 0 & 1 \\ v_2 & 1 & 0 & 1 & 0 & 1 & 0 \\ v_3 & 0 & 1 & 1 & 1 & 0 & 0 \\ v_4 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Table 7: Incidence matrix of undirected graph A

Do note: This only makes sense with numbered edges.

By looking at a column, you can see which vertices are connected to that edge. Incidence matrices can also represent multiple edges by having multiple of the same column, and loops by having a column with only one 1 in it.

10.6 – Isomorphism

An isomorphism is a one-to-one function between the vertex sets of two graphs that preserves adjacency. Two graphs G and H are isomorphic if there exists an isomorphism between them. This means that just by moving vertices and edges around, the two graphs can be made identical.

To determine that two graphs **are not** isomorphic, you can check:

- Different number of vertices
- Different number of edges
- Different degree sequences
- Different properties (e.g. bipartiteness, presence of cycles)
- Different subgraph structures (e.g. one has a complete subgraph that the other lacks)
- Different paths between vertices (e.g. one has a path length 3 passing through vertices of degrees 2,3,2, while the other does not)
- Different connectedness (e.g. one is connected, the other is not, or for directed graphs, one is strongly connected while the other is not)

10.7 – Connectedness

A path is a sequence of edges that connects a sequence of vertices. For simple graphs, they can also be described as a sequence of vertices. Paths can only connect neighbours.

A path is simple if no edges are repeated. A path is a cycle if it starts and ends at the same vertex.

Undirected graphs

An undirected graph is connected if there is a path between every pair of vertices.

A connected component is a maximal connected subgraph of an undirected graph.

Directed graphs

A directed graph is strongly connected if there is a directed path from every vertex to every other vertex.

A directed graph is weakly connected if the underlying undirected graph is connected.

A strong component is a maximal strongly connected subgraph of a directed graph.

10.8 – Disconnecting graphs

A *cut-vertex* is a vertex that, when removed along with its incident edges, increases the number of connected components of the graph. Graphs without cut-vertices are called nonseparable.

A *cut-edge* (or *bridge*) is an edge that, when removed, increases the number of connected components of the graph.

A *vertex cut* or *separating set* is a set of vertices that, when removed along with their incident edges, increases the number of connected components of the graph.

An *edge cut* is a set of edges that, when removed, increases the number of connected components of the graph.

Relation

The *connectivity* $\kappa(G)$ of a graph G is the size of the smallest vertex cut.

The *edge-connectivity* $\lambda(G)$ of a graph G is the size of the smallest edge cut.

$$\kappa(G) \leq \lambda(G) \leq \min_{v \in V} (\deg(v))$$

Only complete graphs have $\kappa(G) = \lambda(G) = \min_{v \in V} (\deg(v)) = n - 1$.

Only disconnected graphs have $\kappa(G) = \lambda(G) = \min_{v \in V} (\deg(v)) = 0$.

10.9 – Amount of paths between vertices and shortest distance

Given adjacency matrix A of graph G , the entry a_{ij} in matrix A^k gives the amount of different paths of length k from vertex i to vertex j .

The shortest distance $d(u, v)$ between vertices u and v is the smallest value of k for which $a_{uv} \neq 0$ in A^k .

10.10 – Euler paths

An *Euler path* is a path that uses every edge in a graph exactly once.

An *Euler circuit* is an Euler path that starts and ends at the same vertex.

A connected multigraph has an Euler circuit $\iff$ all vertices have even degree.

A connected multigraph has an Euler path $\iff$ exactly two vertices have odd degree.

10.11 – Hamilton Paths

A *Hamilton path* is a path that visits every vertex in a graph exactly once.

A *Hamilton circuit* is a Hamilton path that starts and ends at the same vertex.

Dirac's Theorem: If G is a simple graph with n vertices ($n \geq 3$) such that the degree of every vertex in G is at least $n/2$, then G has a Hamilton circuit.

Ore's Theorem: If G is a simple graph with n vertices ($n \geq 3$) such that for every pair of non-adjacent vertices u and v , $\deg(u) + \deg(v) \geq n$, then G has a Hamilton circuit.

If a graph has a vertex of degree 1, it cannot have a Hamilton circuit.

2 edges of every vertex must be part of any Hamilton circuit.

Complete graphs K_n with $n \geq 3$ have Hamilton circuits.

11 Polynomials

An n 'th degree polynomial can be written as $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$ where $a_n \neq 0$. The degree of the polynomial is n .

$[x^k]P(x)$ denotes the k 'th degree coefficient. So for $P(x) = 3x^2 + 4x + 2$ then $[x^1]P(x) = 4$

Algebra

Addition:

$$(ax^2 + bx + c) + (dx^2 + ex + f) = (a + d)x^2 + (b + e)x + (c + f)$$

Which means that:

$$[x^k](P(x) + Q(x)) = [x^k]P(x) + [x^k]Q(x)$$

Scalar multiplication:

$$c \cdot 4x^2 + 7 = (c \cdot 4)x^2 + (c \cdot 7)$$

Which means that:

$$[x^k](cP(x)) = c[x^k]$$

Multiplication: For $P(x) = ax^3 + bx$ and $Q(x) = cx^3 + dx^2 + ex + f$:

$$P(x)Q(x) = (ac)x^6 + (ad)x^5 + (ae + bc)x^4 + (af + bd)x^3 + (be)x^2 + (bf)x$$

Which means that:

$$[x^k](P(x)Q(x)) = \sum_{k=0}^n [x^k]P(x) \cdot [x^{n-k}]Q(x)$$

11.1 – Other info

The division algorithm for polynomials states that for two polynomials $P(x)$ and $D(x)$, there exist unique polynomials $Q(x)$ (the quotient) and $R(x)$ (the remainder) such that:

$$P(x) = D(x)Q(x) + R(x)$$

where $\deg(R(x)) < \deg(D(x))$ or $R(x) = 0$.

Theorem 6.1